

SuratJalan.AI — COMPFEST 18 AIC Submission & Deliverables Checklist

Competition: COMPFEST 18 AI Innovation Challenge (AIC) — Fasilkom Universitas Indonesia

Theme: AI for the Backbone of the Economy

Pillar: Smart Logistics (Gudang, Distribusi & Pergerakan Barang)

Innovation Title: **SuratJalan.AI (ResiVision)** — AI-Powered Proof-of-Delivery Audit & Instant Invoice Reconciliation Engine for Indonesian Supply Chains

Institution: Universitas Bina Nusantara (BINUS University)

Ringkasan Luaran dan Bobot Penilaian

Deliverable Item	Scoring Weight	Target Format	Public / Access
1. Public GitHub Repository	25% (Tech & Architecture) + 15% (MVP Readiness)	GitHub Repo URL	Public
2. Proposal Document	15% (Proposal) + 20% (Originality) + 10% (Theme)	PDF (Max 20 pages)	Direct File / Cloud Link
3. Video Proof of Work (PoW)	Part of Technical Evaluation	YouTube Video (Max 7 mins)	Unlisted
4. Video Promosi Inovasi	15% (Promotion Video)	YouTube Video (Max 5 mins)	Public
5. Bonus Points Package	+5.0% Bonus	Proposal & UI Sections	Included in Docs & App

1. Public GitHub Repository Checklist (Weight: 25% + 15%)

- [x] **Standalone Public Repository:**
- Repo URL: `https://github.com/hi-aprilwang/suratjalan-ai`
- [x] **0-Config Local Reproducibility:**
- [x] `docker-compose.yml` configured for 1-command startup (`docker compose up --build`)
- [x] Offline high-fidelity deterministic fallback engine (app runs 100% without external API keys)
- [x] Clear setup instructions for both Docker and local development in `README.md`
- [x] **Modular Architecture & Clean Code:**
- [x] Frontend (`frontend/`): Next.js 16 + React 19 + TypeScript + Tailwind CSS
- [x] Backend (`backend/`): FastAPI + Pydantic v2 + Pillow
- [x] AI Engine (`backend/app/services/`): Google Gemini 2.0 Flash VLM with spatial bounding-box grounding
- [x] Synthetic Generator (`synthetic_generator/`): Indonesian Surat Jalan document synthesis pipeline
- [x] **Git Hygiene & Conventional Commits:**
- [x] Commits follow standard format: `feat:`, `fix:`, `refactor:`, `docs:`, `chore:`
- [x] Clean `.gitignore` (no `.env`, `node_modules`, `venv`, or `.pyc` tracked)
- [x] **Pre-loaded Indonesian Presets (6 Core Enterprise Scenarios):**
- [x] Preset 1: PT Indofood CBP Sukses Makmur (100% Matched -> Approved)
- [x] Preset 2: PT Mayora Indah (8 Wet Cartons Returned -> Flagged Claim IDR 1,440,000)

- [x] Preset 3: PT Sayap Mas Utama / Wings Group (Damaged Items + Missing Stamp -> Rejected Claim IDR 2,780,000)
- [x] Preset 4: PT Frisian Flag Indonesia (Cold Chain Temp Abuse -> Flagged Claim IDR 3,300,000)
- [x] Preset 5: PT Semen Indonesia (20 Rain Damaged Cement Sacks -> Flagged Claim IDR 1,360,000)
- [x] Preset 6: PT Kalbe Farma (Pharma CDOB Expiry Rejection -> Rejected Claim IDR 27,000,000)
- [x] **Enterprise ERP Export Integrations:**
- [x] SAP S/4HANA BAPI JSON payload generator
- [x] Odoo ERP Stock Picking payload generator
- [x] Jurnal.id Auto Debit Memo integration

2. Proposal Document Checklist (Weight: 45% Combined)

- [x] **Document Format & Restrictions:**
- [x] Format: Single PDF file (docs/submission/PROPOSAL_INOVASI.pdf & docs/PROPOSAL LENGKAP_COMPFEST_18.md)
- [x] Page Limit: Compliant with maximum **20 pages** (excluding Cover Page and Table of Contents)
- [x] Language: Bahasa Indonesia (professional academic & business tone)
- [x] **Section 1: Identitas & Executive Summary:**
- [x] Data Anggota Tim (Bina Nusantara University: Azka, Ghaus, JW, Daniel, April)
- [x] Judul Inovasi & Tagline (*Surat Jalan.AI — #EncloseTheGap*)
- [x] Executive Summary (1–page snapshot of the Indonesian supply chain crisis & SuratJalan.AI solution)
- [x] **Section 2: Latar Belakang & Urgensi (Indonesian Problem Context):**
- [x] The 3–ply carbon paper *Surat Jalan* bottleneck in Indonesia
- [x] 14–30 day invoice delays causing MSME working capital freeze
- [x] FMCG DC check-in friction (Indomaret DC, Alfamart DC, Hypermart)
- [x] **Section 3: Solusi & Orisinalitas Inovasi:**
- [x] Multimodal VLM with spatial coordinate grounding ([ymin, xmin, ymax, xmax])
- [x] Handwritten strikethrough detection + physical warehouse stamp verification
- [x] Automatic IDR claim calculation vs Purchase Order baseline
- [x] **Section 4: Metodologi & Arsitektur Teknis:**
- [x] System Architecture Diagram (FE, BE, Gemini VLM, ERP Dispatcher)
- [x] Data pipeline & synthetic data augmentation methodology
- [x] Discrepancy math & deterministic validation logic
- [x] **Section 5: Analisis Kelayakan, Model Bisnis & Unit Economics (+3.5% Bonus):**
- [x] Unit Economics Table (Cost per document: Rp 2.4 vs Rp 3,500 manual entry)
- [x] B2B SaaS pricing model (Tiered per-audit or enterprise volume subscription)
- [x] Target market sizing (3PL freight carriers, FMCG principals, distributor networks)
- [x] **Section 6: Tata Kelola AI Etis, Privasi & Keamanan (AI Governance Bonus):**
- [x] Data privacy: Enterprise PII redaction and secure in-memory processing
- [x] Human-in-the-loop (HITL) audit verification for high-risk claims
- [x] Model bias mitigation across diverse handwriting and physical stamp ink conditions

3. Video Proof of Work (PoW) Checklist (Unlisted YouTube)

- [] **Format & Specifications:**
- [] Maximum Duration: **7 minutes**
- [] Privacy Setting: **Unlisted** (bisa diakses via link)
- [] Title Format: `COMPFEST 18 AIC: PROOF OF WORK - [Nama Tim] - SuratJalan.AI`

- ☐ **STRICT RULE: NO CUTS / EDIT CUTS** during the technical execution sequence (Fast-forwarding & voiceover / narration are allowed).
- ☐ **Screen Layout & Visual Elements:**
- ☐ Split screen or visible layout showing:
 - ☐ Terminal running Docker / startup commands and live backend logs
 - ☐ Web Browser workstation UI (`http://localhost:3000`)
 - ☐ **Visible Real-Time System Clock** (Windows taskbar clock / digital widget)
- ☐ **Demonstration Flow:**
- ☐ 1. Clone repo & launch via `docker compose up --build` (or start backend & frontend)
- ☐ 2. Open `http://localhost:3000` on browser
- ☐ 3. Run Preset 1 (Indofood Clean Match -> Approved)
- ☐ 4. Run Preset 2 (Mayora Return Discrepancy -> Show spatial bounding boxes on canvas & claim math)
- ☐ 5. Run Preset 3 (Wings Group Missing Stamp -> Rejected alert)
- ☐ 6. Trigger ERP Export modal (demonstrate SAP / Odoo / Jurnal.id JSON payload dispatch)

4. Video Promosi Inovasi Checklist (Public YouTube, Max 5 Mins)

- ☐ **Format & Specifications:**
- ☐ Maximum Duration: **5 minutes**
- ☐ Privacy Setting: **Public**
- ☐ Title Format: `COMPFEST 18 AIC: [Nama Tim] - SuratJalan.AI`
- ☐ Description: Include brief project summary, team members, and GitHub repository link
- ☐ **Content & Storyboard Structure:**
- ☐ **0:00 – 1:00 (The Hook & Indonesian Problem):**
 - Indonesian logistics reality: delivery trucks queuing at DCs, stacks of crumpled carbon *Surat Jalan*, 30-day invoice delays.
- ☐ **1:00 – 2:30 (Introducing SuratJalan.AI & Solution):**
 - What is SuratJalan.AI? Explain Gemini 2.0 Flash VLM + Spatial Grounding + Automated Reconciliation.
- ☐ **2:30 – 3:45 (Product Walkthrough / UI Highlights):**
 - Crisp showcase of the interactive workstation UI, hoverable bounding boxes, instant discrepancy claims.
- ☐ **3:45 – 4:30 (Business Value & Economic Impact):**
 - 99.9% cost reduction (Rp 2.4/doc), instant cash flow clearance for Indonesian MSMEs.
- ☐ **4:30 – 5:00 (Vision & Closing Call-to-Action):**
 - Aligned with COMPFEST 18 theme: *AI for the Backbone of the Economy* `#EncloseTheGap`.

5. Bonus Points Checklist (+5.0% Total)

- ☒ **Realistic Unit Economics & ROI Impact (+3.5%):**
 - Documented in Proposal & `README.md` comparing manual audit vs AI audit.
 - ☒ **Responsible AI & Governance (+3.5%):**
 - Documented in Proposal & Architecture docs (HITL, explainability with spatial bounding boxes).
 - ☐ **AIC Talks Attendance (+1.5%):**
 - Team presence confirmed during official COMPFEST 18 AIC Talks webinar session.
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6. Final Pre-Submission Validation

- ☐ Verify GitHub repository is set to **Public** and accessible without login.
- ☐ Test YouTube video links in an incognito / private browser window:
- ☐ Proof of Work video link (Unlisted) plays properly with clear audio.
- ☐ Promosi video link (Public) is live.
- ☐ Ensure Proposal PDF file size complies with submission upload limits (< 20MB).
- ☐ Double-check all team member information (Nama, NIM/NIK, Email, No. HP, Universitas) on the official COMPFEST submission portal.